

Daily Solar Energy Wastage Prediction using Machine Learning (Sri Lanka Context)

1. Project Overview

Sri Lanka has seen a rapid increase in household rooftop solar installations. During sunny days, especially around midday, many households export excess solar energy to the electricity grid (CEB/LECO). However, due to grid absorption limitations and low demand during these hours, not all exported solar energy can be utilized. This results in energy wastage (curtailment) at the grid level.

The objective of this Machine Learning (ML) project is to predict the **daily amount of solar energy wasted after export to the grid**, using historical solar generation, household load patterns, and Sri Lanka-specific weather and solar irradiance data.

2. Problem Definition

Goal: Predict the daily solar energy wastage (in kWh) that occurs after households export excess solar energy to the grid.

ML Task Type: Regression

Target Variable (Dependent Variable): `Daily_Wasted_Energy_kWh`

3. Data Sources (Confirmed & Public)

3.1 Solar Generation and Household Load Data

These datasets provide real household-level time-series data for solar PV generation and electricity consumption.

- **Photovoltaic Generation and Load Demand Dataset (Kaggle)**

<https://www.kaggle.com/datasets/pythonaifroz/photovoltaic-generation-and-load-demand-datasets>

- **Household Load and Solar Generation Dataset (Kaggle)**

<https://www.kaggle.com/datasets/mexwell/household-load-and-solar-generation>

These datasets typically include: - Date & time - Solar PV generation (kWh or kW) - Household electricity consumption (load)

3.2 Sri Lanka Weather & Solar Irradiance Data (Area-wise)

This data is used to represent local solar conditions for **Colombo, Galle, and Matara**.

- **NASA POWER – Solar & Meteorological Data**
<https://power.larc.nasa.gov/data-access-viewer/>

From this source, collect **daily data** for each location: - Global Horizontal Irradiance (GHI) - Average temperature

Locations: - Colombo - Galle - Matara

4. Independent Variables (X)

The following features are used as inputs to the ML models:

Time-based Features

- Day of week
- Month
- Is weekend (Yes/No)

Area

- Area (Colombo / Galle / Matara)

Weather & Solar Conditions

- Daily solar irradiance (GHI)
- Average temperature

Energy Behaviour

- Daily solar PV generation (kWh)
 - Daily household electricity load (kWh)
 - Previous day solar generation (kWh)
 - Previous day household load (kWh)
-

5. Dependent Variable (Y)

Daily Wasted Energy Calculation

The daily wasted energy is computed using the following logic:

1. Exported Energy: $\text{Exported_Energy} = \text{Daily_PV_Generation} - \text{Daily_Household_Load}$

2. Grid Export Limit (Assumption): $\text{Export_Limit} = 10 \text{ kWh per day}$ (Represents grid absorption limitations during high solar penetration)

3. Daily Wasted Energy:

4. If $\text{Exported_Energy} \leq 0 \rightarrow \text{Wasted_Energy} = 0$

5. If $\text{Exported_Energy} > 0 \rightarrow \text{Wasted_Energy} = \max(0, \text{Exported_Energy} - \text{Export_Limit})$

This calculated value becomes the dependent variable: `Daily_Wasted_Energy_kwh`

6. Data Preparation Steps

1. Clean missing or inconsistent values
 2. Convert hourly data to daily totals (if required)
 3. Merge PV/load data with weather data by date and area
 4. Create lag features (previous day values)
 5. Encode categorical variables
 6. Normalize numeric features (optional)
-

7. Machine Learning Models Used

The following regression algorithms will be trained and compared:

1. Linear Regression
 2. Ridge Regression
 3. Decision Tree Regressor
 4. Random Forest Regressor
 5. K-Nearest Neighbors (KNN) Regressor
 6. Gradient Boosting Regressor
 7. Support Vector Regressor (Optional)
-

8. Model Evaluation Metrics

Each model is evaluated using: - Mean Absolute Error (MAE) - Root Mean Squared Error (RMSE) - R^2 Score

The best-performing model is selected based on the lowest RMSE.

9. Web Dashboard (ML Coursework Demonstration)

A simple web dashboard will be developed using **Streamlit** to present results.

Dashboard Features: - Dataset overview - Model comparison table - Actual vs predicted wasted energy chart
- Area selection (Colombo / Galle / Matara) - Input form for prediction

10. Conclusion

This project demonstrates how machine learning can be used to analyze and predict solar energy wastage due to grid management limitations. By leveraging publicly available datasets and Sri Lanka-specific solar data, the system provides insights that can support better solar energy utilization planning.

11. Tools & Technologies

- Python
- Pandas, NumPy
- Scikit-learn
- Streamlit
- Kaggle Datasets
- NASA POWER Data